1. Advanced Log Analysis

Objective

To analyze logs, find suspicious activity, and improve detection using Elastic Security.

Log Correlation

I uploaded sample logs into Kibana and filtered Event ID 4625 (failed login). Then I checked for outbound traffic using destination IP. I compared timestamps and found suspicious activity from the same IP.

Table

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00	4625	192.168.1.100	8.8.8.8	Suspicious

Anomaly Detection

I created a rule in Elastic to detect high data transfer using:
network.bytes_out > 1000000
The rule triggered when large data was sent.

Log Enrichment

I used GeoIP to add location to IP addresses. This helped identify where traffic was coming from.

Summary (50 words)

GeoIP enrichment adds location details to IP addresses. This helps identify suspicious activity from unknown regions. It improves analysis and makes threat detection faster and easier for security analysts.

2. Threat Intelligence Integration

Objective

To use threat intelligence to detect malicious activity.

Threat Feed Import

I connected AlienVault OTX with Wazuh using API key. I tested with IP 192.168.1.100 and observed alerts.

Alert Enrichment

I checked alert details and found reputation data from OTX.

Table

Alert ID	IP	Reputation	Notes
003	192.168.1.100	Malicious	C2 activity

Threat Hunting

I searched logs using:
user.name != "system"
This helped identify suspicious user activity.

Summary (50 words)

Threat hunting helps detect misuse of valid accounts. By filtering non-system users, unusual login activity can be found. This improves detection of compromised accounts and insider threats.

3. Incident Escalation Practice

Objective

To handle incidents and escalate them properly.

Escalation Simulation

I created a case in TheHive for unauthorized access and marked it as high priority. Then I escalated it to Tier 2.

Summary (100 words)

A high priority incident was detected on Server-Y at 13:00. The IP 192.168.1.200 was involved. It matched MITRE T1078 (valid accounts). The system was isolated to prevent damage. Logs were collected for analysis. The issue was serious, so it was escalated to Tier 2 for further investigation.

SITREP

Title: Unauthorized Access on Server-Y

Summary: Detected at 13:00 from IP 192.168.1.200.

Actions:

- Server isolated
- IP blocked
- Escalated to Tier 2

Workflow Automation

I created a playbook in Splunk Phantom. It checks if severity is high and assigns the case to Tier 2 automatically.

Conclusion

This practical helped me understand log analysis, threat intelligence, and incident response. These skills are important for working in a SOC environment.

4. Post-Incident Analysis

Objective

To analyze a security incident, identify root cause, and measure SOC performance.

Root Cause Analysis (5 Whys)

Aim

To identify the root cause of a phishing incident.

Table

Question	Answer
Why was email opened?	User clicked malicious link
Why was link clicked?	Weak email filtering
Why was filtering weak?	Outdated security rules
Why were rules outdated?	No regular updates

Question	Answer
Why no updates?	Lack of monitoring process

Result

The root cause was lack of proper security monitoring and outdated filtering rules.

Fishbone Diagram

Aim

To identify contributing factors of the phishing incident.

Causes Identified

- People: Lack of awareness
- Process: No regular security checks
- Technology: Weak email filter
- Environment: External phishing attack

(Use Draw.io to create diagram and upload screenshot separately)

Metrics Calculation

Aim

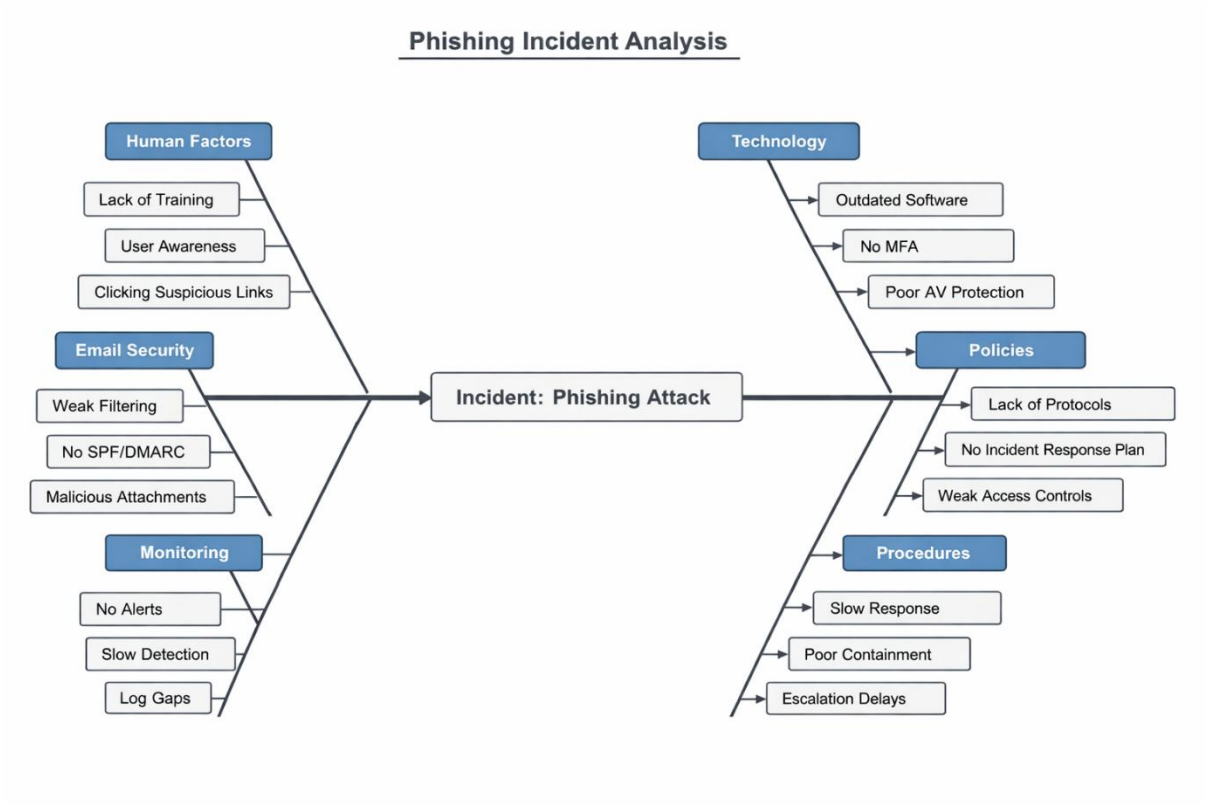
To measure SOC efficiency using MTTD and MTTR

Values

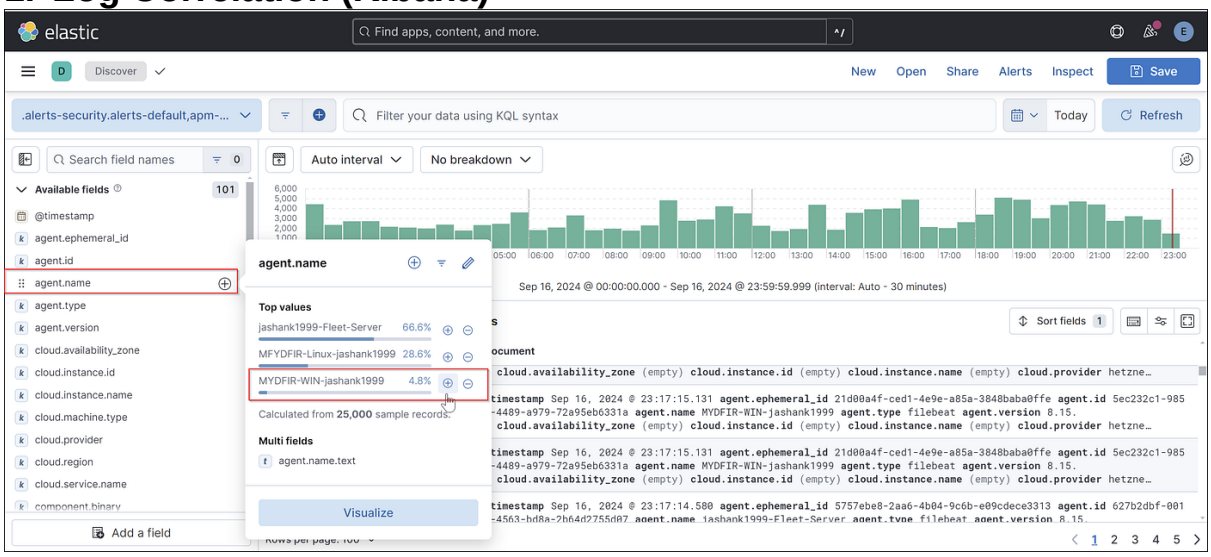
- MTTD (Mean Time to Detect): 2 hours
- MTTR (Mean Time to Respond): 4 hours

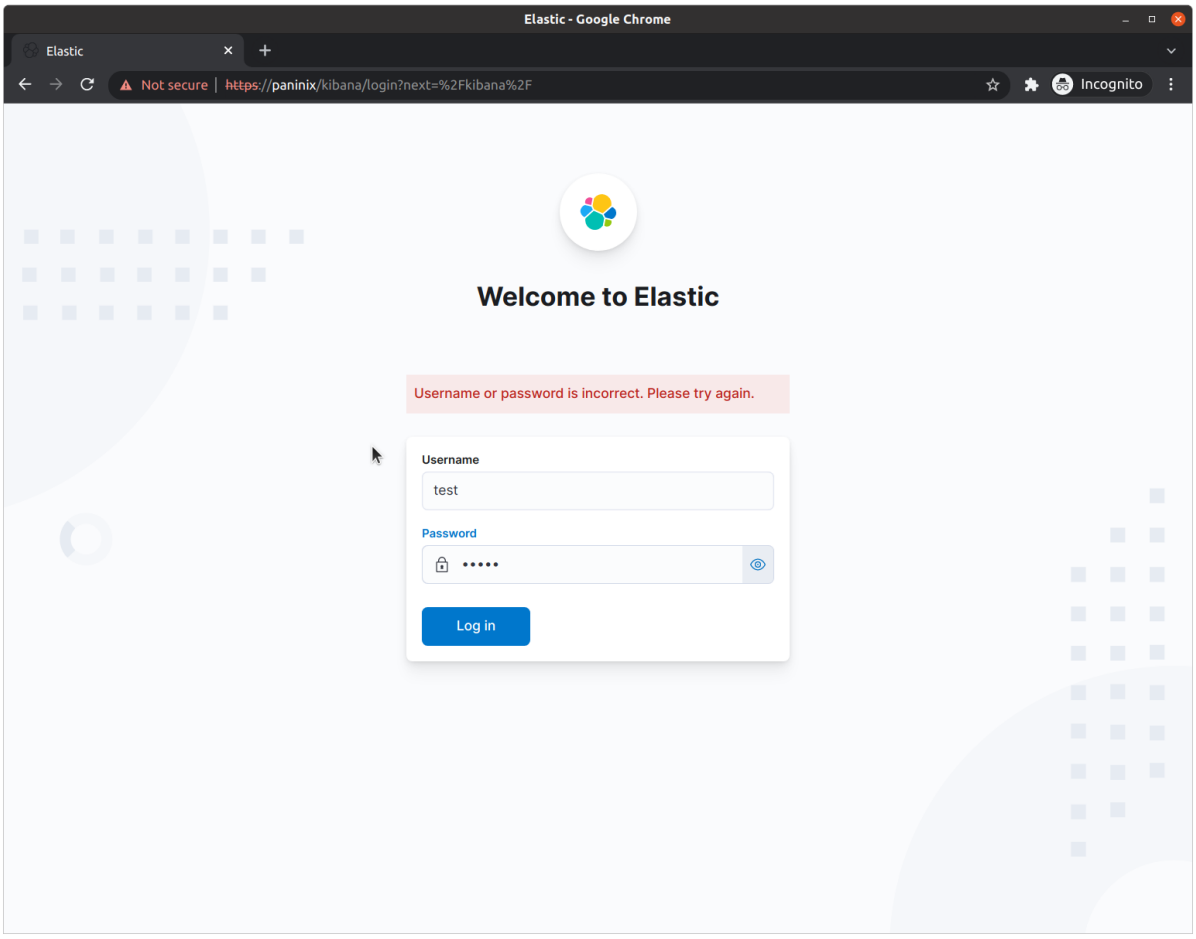
Summary (50 words)

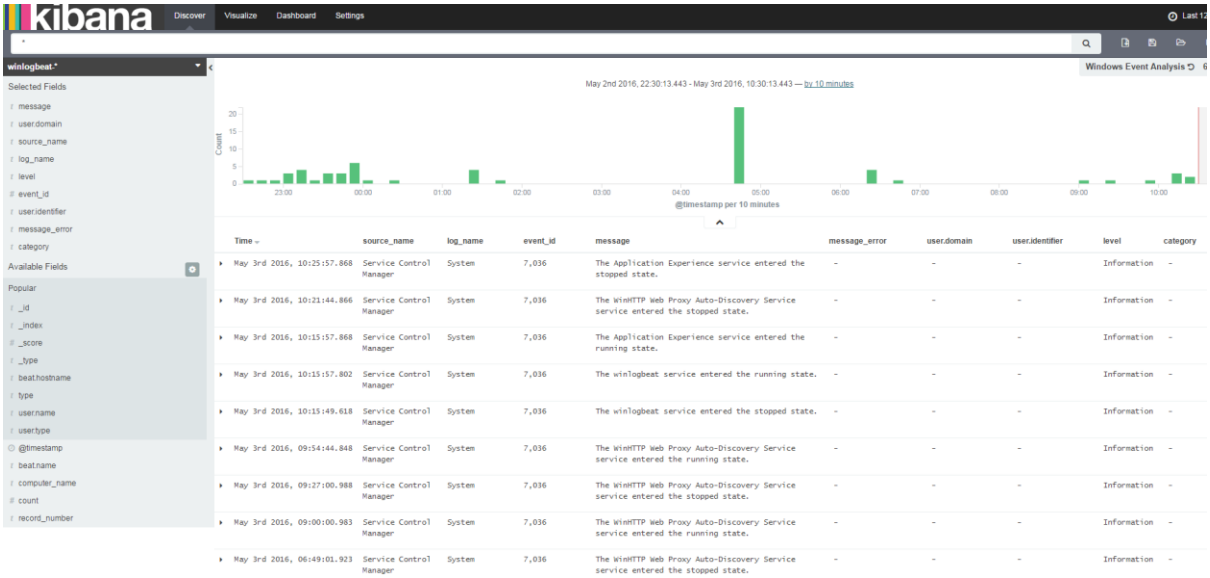
MTTD and MTTR are important SOC metrics used to evaluate performance. In this case, detection took 2 hours and response took 4 hours. Lower values indicate better efficiency. Improving monitoring systems and response processes can reduce these times and strengthen overall incident handling capability.

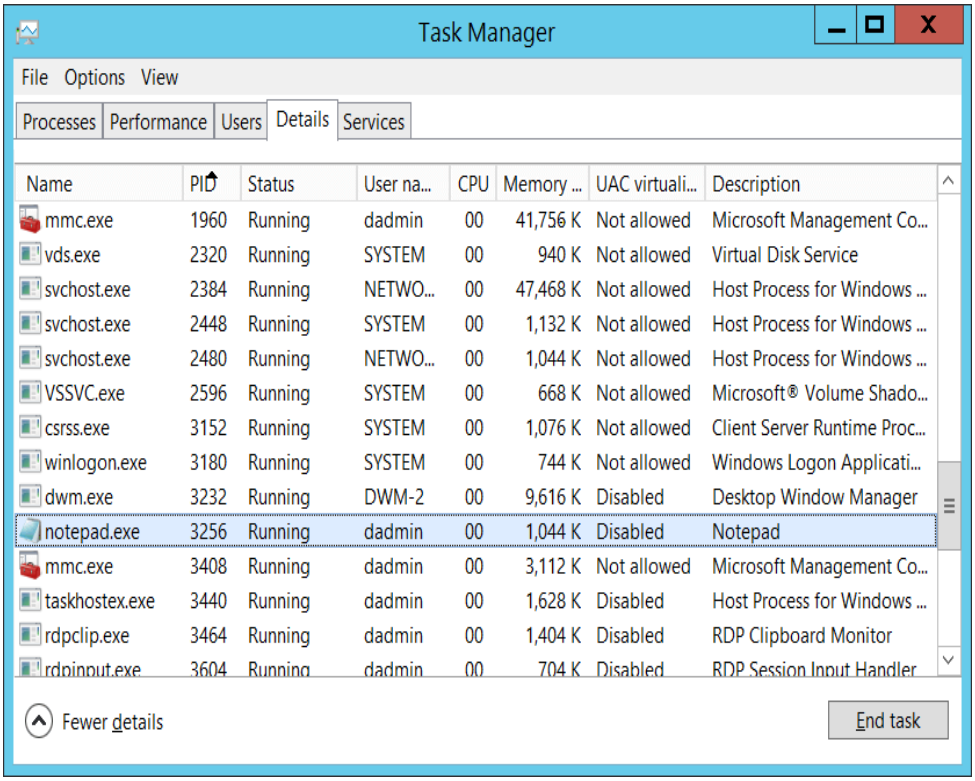


1. Log Correlation (Kibana)

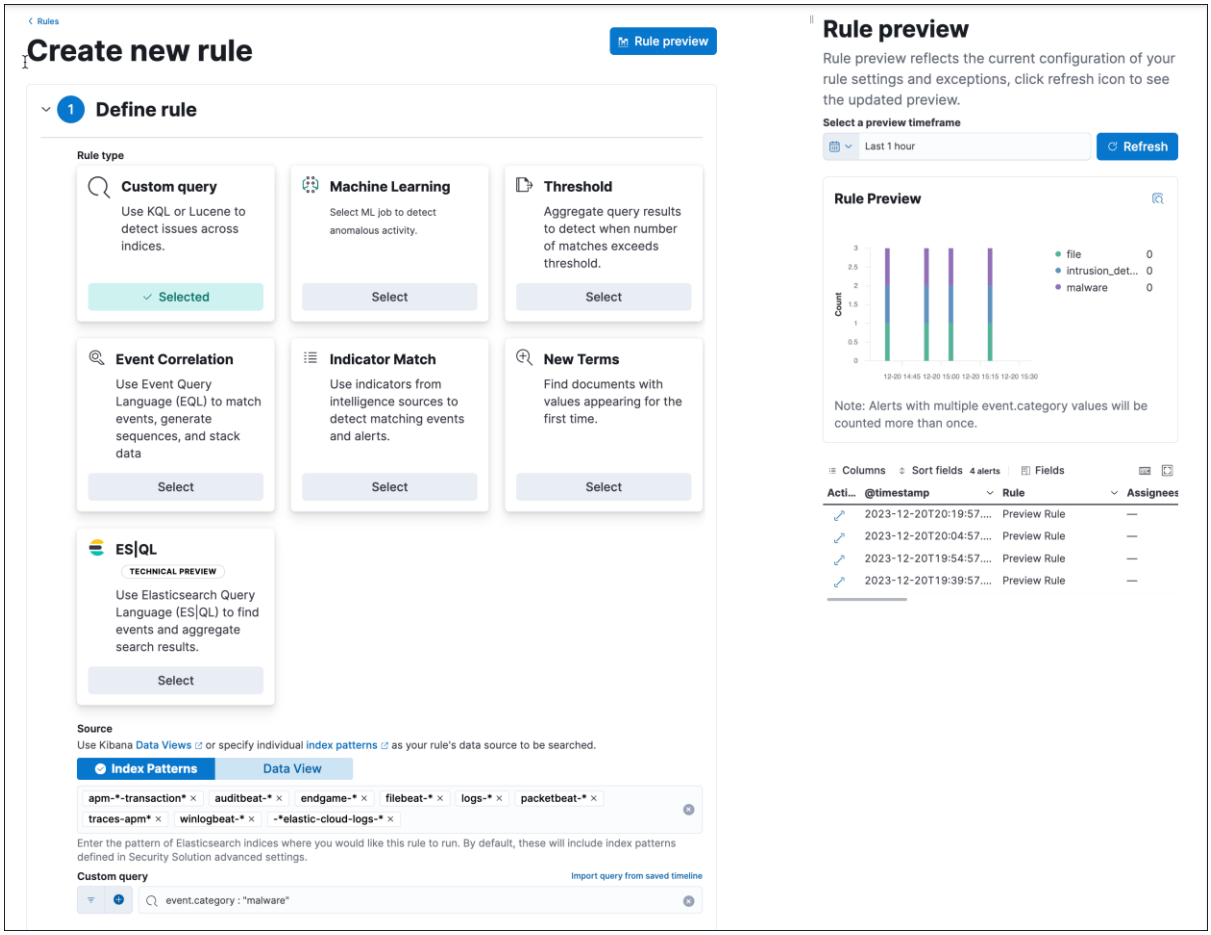


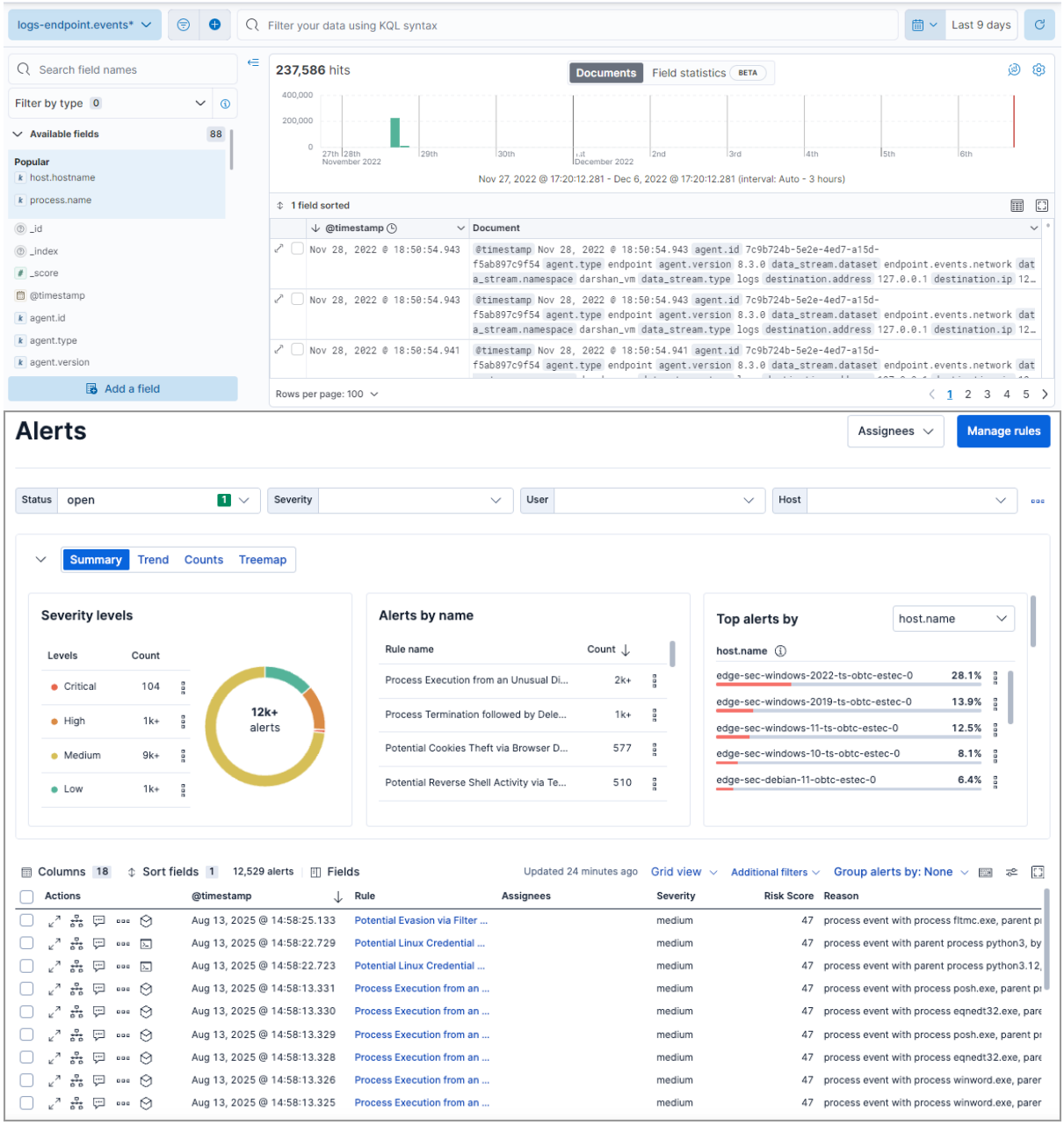

Elastic - Google Chrome
Elastic
Not secure | https://paninix/kibana/login?next=%2Fkibana%2F
Incognito
Welcome to Elastic
Username or password is incorrect. Please try again.
Username
test
Password
Log in


kibana
Discover
Visualize
Dashboard
Settings
winlogbeat.*
Selected Fields
message
user.domain
source_name
log_name
level
event_id
user.identifier
message_error
category
Available Fields
Popular
_id
_index
_score
_type
beat.hostname
type
user.name
user.type
@timestamp
beat.name
computer_name
count
record_number
Windows Event Analysis
May 2nd 2016, 22:30:13.443 - May 3rd 2016, 10:30:13.443 — by 10 minutes
Count
@timestamp per 10 minutes
Time
source_name
log_name
event_id
message
message_error
user.domain
user.identifier
level
category
May 3rd 2016, 10:25:57.868 Service Control Manager System 7,036 The Application Experience service entered the stopped state. - - - Information -
May 3rd 2016, 10:21:44.866 Service Control Manager System 7,036 The WinHTTP Web Proxy Auto-Discovery Service service entered the stopped state. - - - Information -
May 3rd 2016, 10:15:57.868 Service Control Manager System 7,036 The Application Experience service entered the running state. - - - Information -
May 3rd 2016, 10:15:57.802 Service Control Manager System 7,036 The winlogbeat service entered the running state. - - - Information -
May 3rd 2016, 10:15:49.618 Service Control Manager System 7,036 The winlogbeat service entered the stopped state. - - - Information -
May 3rd 2016, 09:54:44.848 Service Control Manager System 7,036 The WinHTTP Web Proxy Auto-Discovery Service service entered the running state. - - - Information -
May 3rd 2016, 09:27:00.988 Service Control Manager System 7,036 The WinHTTP Web Proxy Auto-Discovery Service service entered the stopped state. - - - Information -
May 3rd 2016, 09:00:00.983 Service Control Manager System 7,036 The WinHTTP Web Proxy Auto-Discovery Service service entered the running state. - - - Information -
May 3rd 2016, 06:49:01.923 Service Control Manager System 7,036 The WinHTTP Web Proxy Auto-Discovery Service service entered the stopped state. - - - Information -

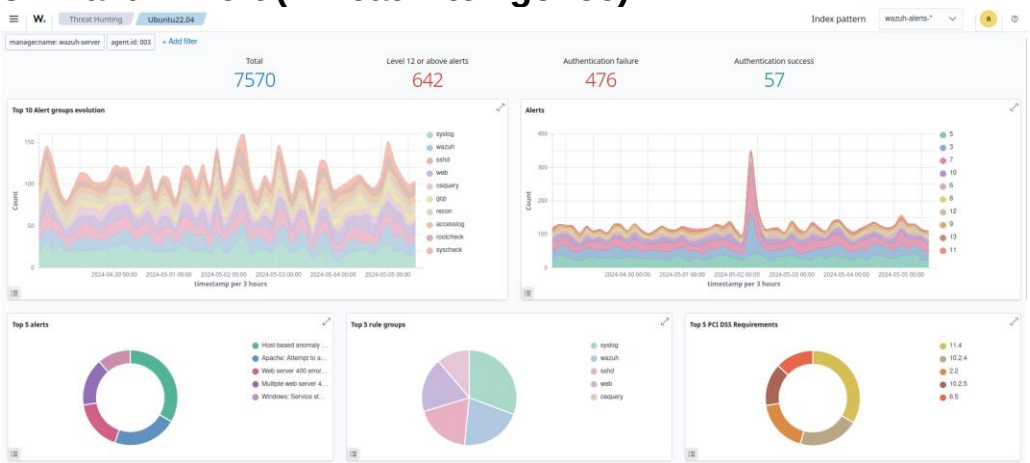


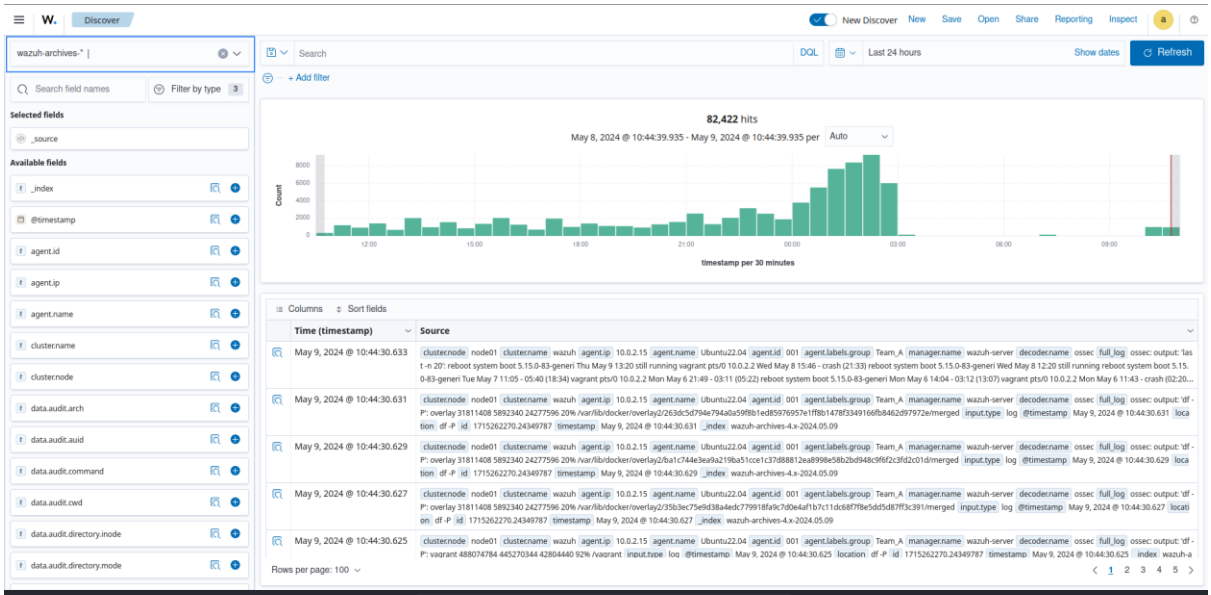
2. Anomaly Detection Rule

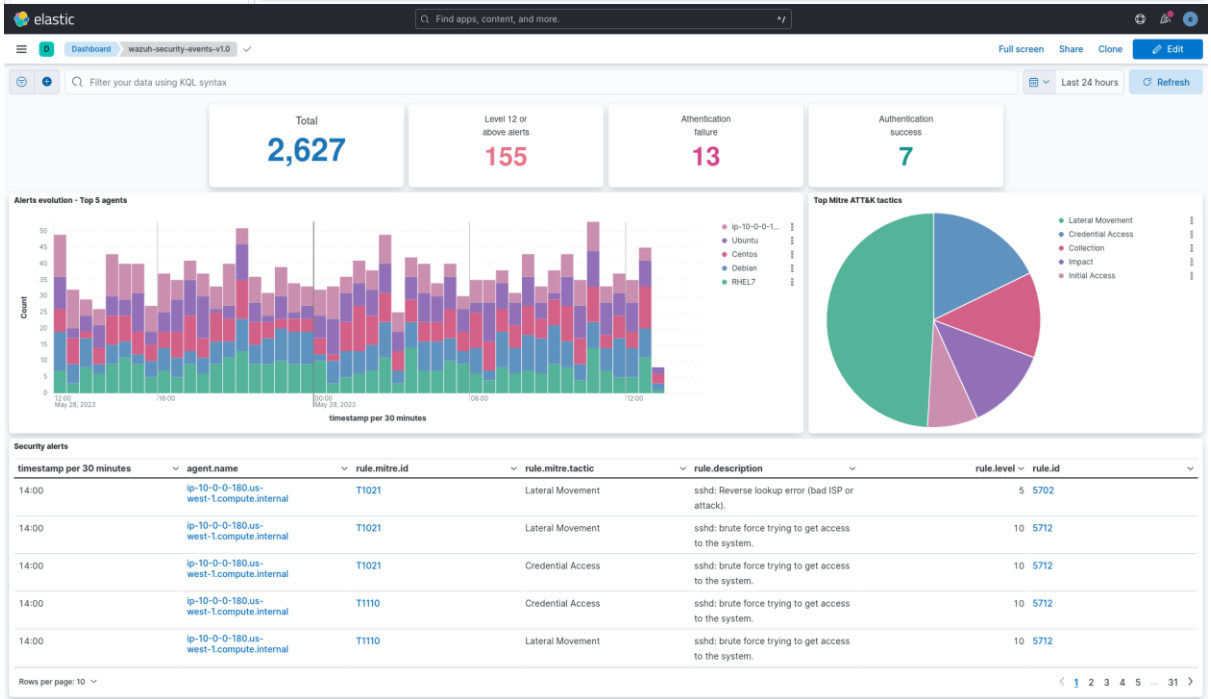




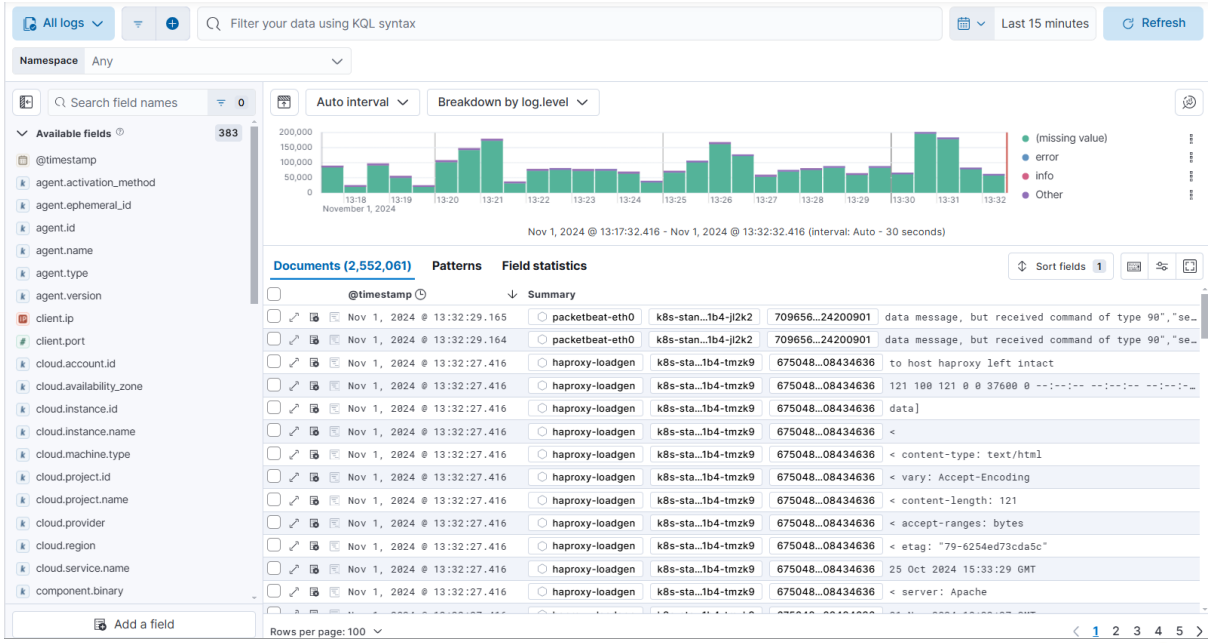
3. Wazuh Alert (Threat Intelligence)

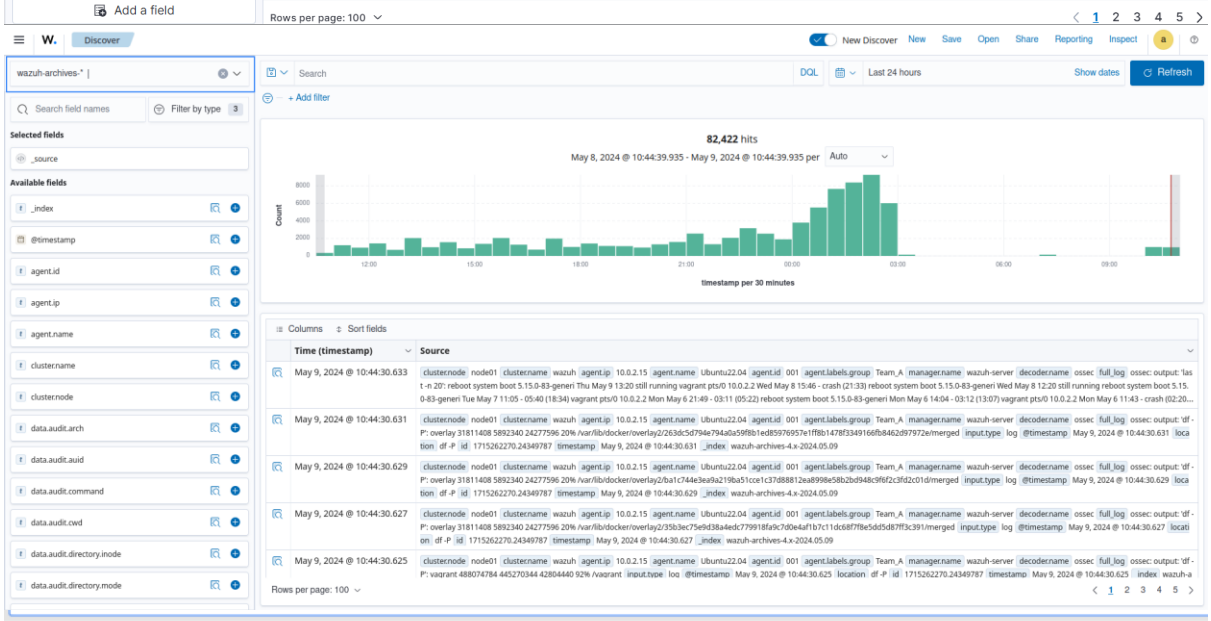






4. Threat Hunting Query





Total Successful Logons

User Name	Domain	Total Logons	Unique Computers	Unique IPs
SYSTEM	NT AUTHORITY	17	1	1
DWM-11	Window Manager	14	1	1
DWM-17	Window Manager	10	1	1
DWM-14	Window Manager	2	1	1
DWM-15	Window Manager	2	1	1
DWM-16	Window Manager	2	1	1
DWM-18	Window Manager	2	1	1
DWM-19	Window Manager	2	1	1
rdc2	PTPLQA	2	1	1